

## Department Overview & History

- **Department Name:** Department of Physics
- **Faculty:** Faculty of Basic Sciences (*historically established under the Faculty of Arts and Basic Sciences*)
- **History:** \* In **2004**, BS Physics and MS Semiconductor Physics were initially launched under the Faculty of Information and Communication Technology (FICT).
  - In **2007**, the Physics department was officially established under the Faculty of Arts and Basic Sciences, and the BS and MS Physics programs were migrated there.
  - In **2017**, the department received a No Objection Certificate (NOC) from the Higher Education Commission (HEC) of Pakistan to offer a PhD program.
- **Academic Standards:** All programs are structured in strict accordance with the HEC guidelines to meet national and international benchmarks.
- **Research & Collaboration:** The department maintains vast inter-departmental and industrial collaborations, encouraging researchers to seek out project opportunities with commercial enterprises.

## 1. Bachelor of Science in Physics (BS Physics)

### Admission Requirements

- **Academic Background:** Intermediate with Physics and Mathematics from any recognized board (or equivalent qualification) with at least **50% marks**.
- **Testing:** Successfully qualifying an aptitude test with a percentage determined by the central Admission Committee.

### Degree Requirements

- **Total Credit Hours:** 137
- **Total Courses:** 42 – 44
- **Minimum Academic Standing:** CGPA  $\geq$  2.2

### Program Educational Objectives (PEOs)

(Note: The webpage template contains placeholder text from a computing discipline template, mapping the following objectives directly to the program):

1. Responsibly practice systematic, disciplined, and quantifiable approaches to solve complex problems in a variety of computing and allied disciplines.
2. Utilize skills and knowledge to serve industry and academia impacting the socio-economic development of Pakistan.
3. Demonstrate sustained learning and adaptation to evolving fields through continued professional development and self-study.

### Curriculum Plan (Program Schema by Semester)

Note: Credit hour listings follow the format (Theory + Lab).

- **1st Semester:** Mechanics (3), Lab-I (0+1), Calculus-I (3), Reading and Writing Skills (3), Pakistan Studies (2), Computer Fundamentals (2+1).
- **2nd Semester:** Electricity and Magnetism-I (3), Heat and Thermodynamics (3), Lab-II

- (0+1), Islamic Studies/Ethics (2), Introduction to Geology (2+1), Calculus-II (3), Communication & Presentation Skills (3).
- **3rd Semester:** Electricity and Magnetism-II (3), Waves and Oscillations (3), Modern Physics (3), Lab-III (0+1), Linear Algebra (3), Academic Reading & Writing (3), Statistics-I (3).
  - **4th Semester:** Optics (3), Lab-IV (0+1), Differential Equations (3), Introduction to Psychology (3), Statistics-II (2+1), Introduction to Chemistry (2+1), Programming Fundamentals (2+1).
  - **5th Semester:** Mathematical Method of Physics-I (3), Electromagnetic Theory-I (3), Classical Mechanics (3), Basic Electronic Devices-I (3), Statistical Physics (3), Lab-V (0+2) (*Prerequisite: Lab-IV*).
  - **6th Semester:** Mathematical Methods of Physics-II (3) (*Prerequisite: MMP-I*), Quantum Mechanics-I (3), Electromagnetic Theory-II (3) (*Prerequisite: EMT-I*), Atomic & Molecular Physics (3) (*Prerequisite: Modern Physics*), Basic Electronic Devices-II (3) (*Prerequisite: BED-I*), Lab-VI (0+2) (*Prerequisite: Lab-V*).
  - **7th Semester:** Quantum Mechanics-II (3) (*Prerequisite: QM-I*), Solid State Physics-I (3), Lab-VII (0+2) (*Prerequisite: Lab-VI*), Project Part-I (0+2), Elective I (3), Elective II (3).
  - **8th Semester:** Solid State Physics-II (3) (*Prerequisite: SSP-I*), Nuclear Physics (3) (*Prerequisite: Modern Physics*), Project Part-II (0+4), Elective III (3), Elective IV (3).

### Available Elective Courses Pool

- PHY-458 Plasma Physics
- PHY-423 Methods of Experimental Physics
- PHY-460 Introduction to Quantum Computing
- PHY-462 Quantum Information Theory
- PHY-461 Quantum Field Theory
- PHY-450 Digital Electronics
- PHY-451 Lasers
- PHY-477 Laser Engineering
- PHY-459 Experimental Techniques in Particle and Nuclear Physics
- PHY-468 Electronic Materials and Devices
- PHY-402 Fluid Dynamics
- PHY-476 Introduction to Photonics
- PHY-454 Introduction to Materials Science
- PHY-455 Introduction to Nano Science and Nanotechnologies
- PHY-493 Particle Physics
- PHY-417 Computer Simulations in Physics
- PHY-494 Surface Sciences
- PHY-416 Computational Physics (*Prerequisite: Programming Fundamentals*)
- PHY-453 Solid State Electronic Devices
- PHY-435 Renewable Energy Resources
- PHY-431 Environmental Physics
- PHY-470 Medical Physics
- PHY-467 Dielectric Materials

- PHY-469 Magnetic Materials
- PHY-478 Quantum Optics
- PHY-424 Experimental Nuclear Physics

## 2. Master of Science in Physics (MS Physics)

### Admission Requirements

- **Academic Background:** 16 years of education in Physics from an HEC-recognized institution with a minimum of **60% marks** under the annual grading system, or a CGPA of **2.7 out of 4.0** under a semester system.
- **Standardized Testing:** Fulfilling one of the following criteria:
  - GAT-General with a minimum 50% score.
  - GAT-Subjective with a minimum 60% score.
  - GRE Subject Test (International) with at least a 50th percentile rank.

### Degree Requirements

- **Total Credit Hours:** 30
- **Total Coursework Requirements:** 8 advanced postgraduate courses (24 credit hours) + Research Thesis (6 credit hours).
- **Minimum Academic Standing:** CGPA  $\geq$  2.5

### Program Schema

- **Semester I (Core Subjects — 12 Credits):** \* PHY-587 Mathematical Methods of Physics (3)
  - PHY-640 Electrodynamics (3)
  - PHY-623 Methods and Techniques of Experimental Physics (3)
  - PHY-677 Quantum Mechanics-I (3)
- **Semester II (Elective Subjects — 12 Credits):** Elective III (3), Elective IV (3), Elective V (3), Elective VI (3).
- **Semesters III & IV (Compulsory Research — 6 Credits):** PHY-667 Research Thesis (6).

## 3. Doctor of Philosophy in Physics (PhD Physics)

### Admission Requirements

- **Academic Background:** An HEC-recognized MS degree or equivalent 18-year master's qualification in Physics. The completed degree must have included a minimum of **30 credit hours** (24 credit hours of graduate level courses + 6 credit hours of thesis) passed with a 1st Division grading or a minimum CGPA of **3.2 out of 4.0**.
- **Standardized Testing:** Prior to admission confirmation, candidates must pass either the standard **GRE International Subject Test** with a minimum 60th percentile rank, or an authorized **GAT Subject Test** with at least **60% marks**.

### Degree Requirements

- **Total Credit Hours:** 18

- **Total Courses Required:** 6 advanced doctoral courses.
- **Examination:** Passing a Comprehensive/DQE (Doctoral Qualification Examination) from the studied coursework.
- **Minimum Academic Standing:** CGPA  $\geq$  3.2